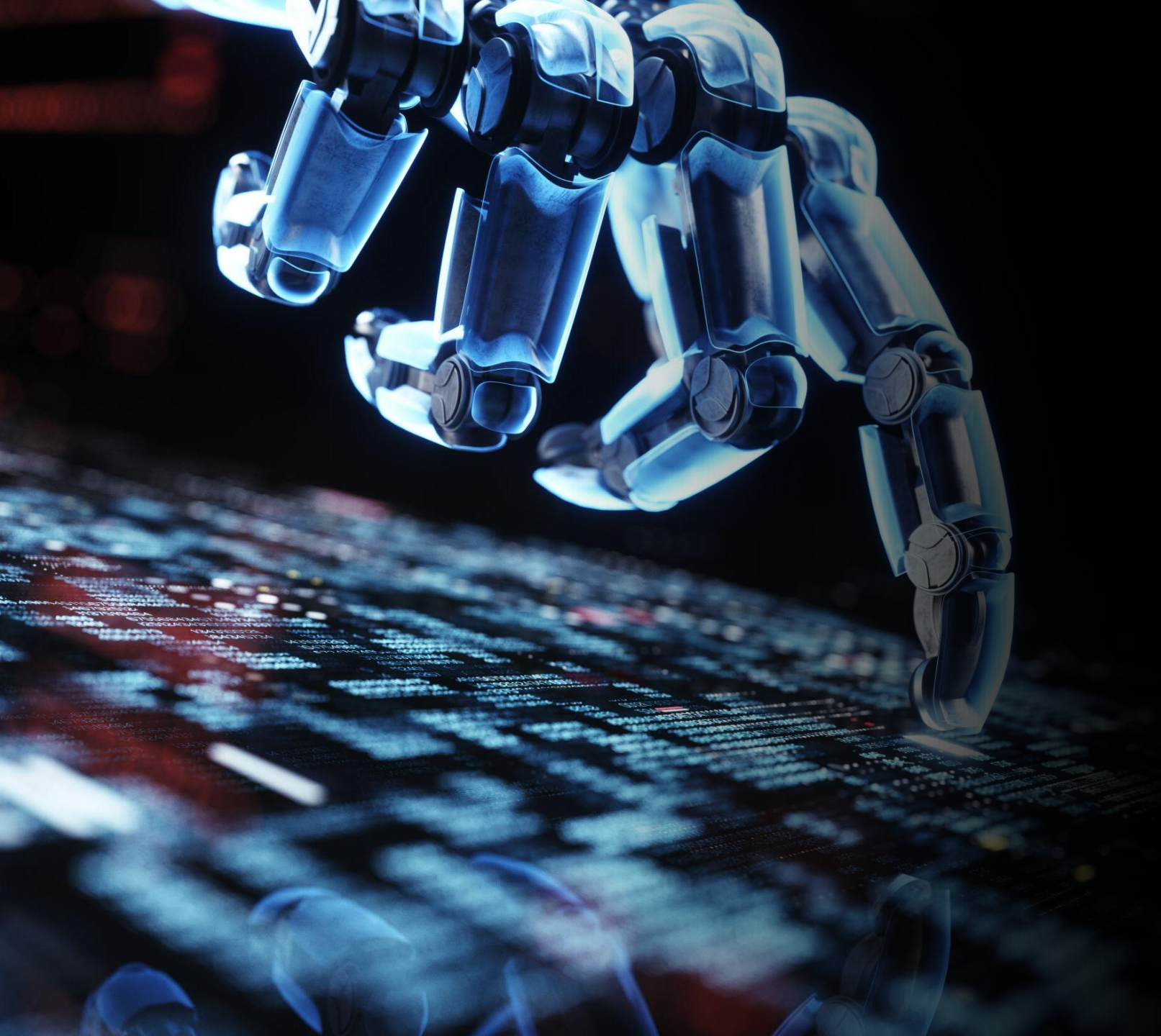




Trajectory Generation

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Reference Book

- Introduction to Robotics- John J. Craig, Third edition, Prentice Hill.

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Mechanical manipulators- introduction

- **Mechanical manipulators** are the most important form of industrial robots.
- Manipulators consist of **nearly rigid links**, which are connected by **joints** that allow relative motion of neighboring links.
- These joints are usually instrumented with **position sensors**, which measure the relative position of neighboring links.
- In the case of **rotary or revolute** joints, these displacements are called **joint angles**.
- Some manipulators contain **sliding (or prismatic) joints**, in which the relative displacement between links is a **translation**, sometimes called the **joint offset**.



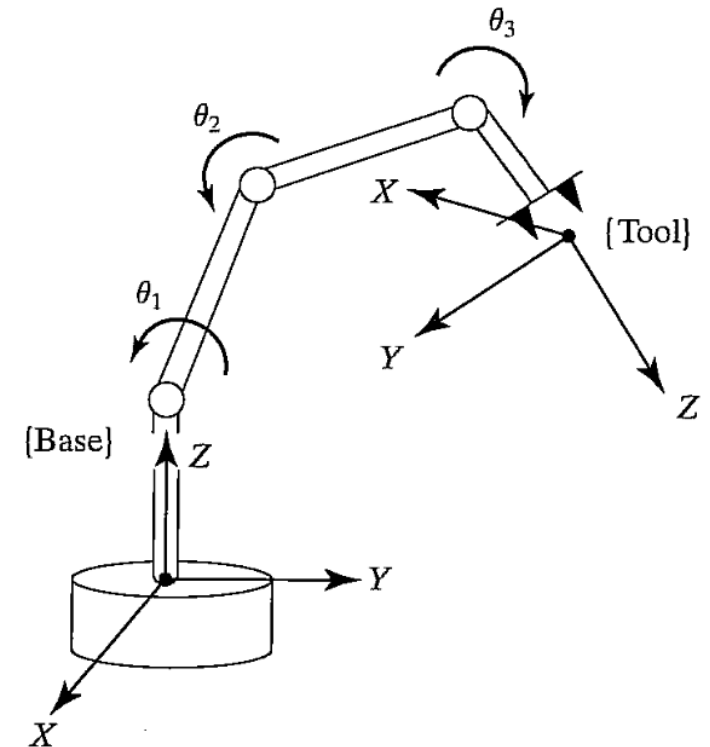
Mechanical manipulators- introduction

- The number of **degrees of freedom** is the number of **independent position variables** needed to locate all parts of the mechanism.
- Because a manipulator is usually **an open kinematic chain**, and because each joint position is usually defined with a single variable, **the number of joints equals the number of degrees of freedom**.
- At the **free end** of the chain of links that make up the manipulator is the **end-effector**.
- Depending on the intended application of the robot, the **end-effector could be** a gripper, a welding torch, an electromagnet, or another device.



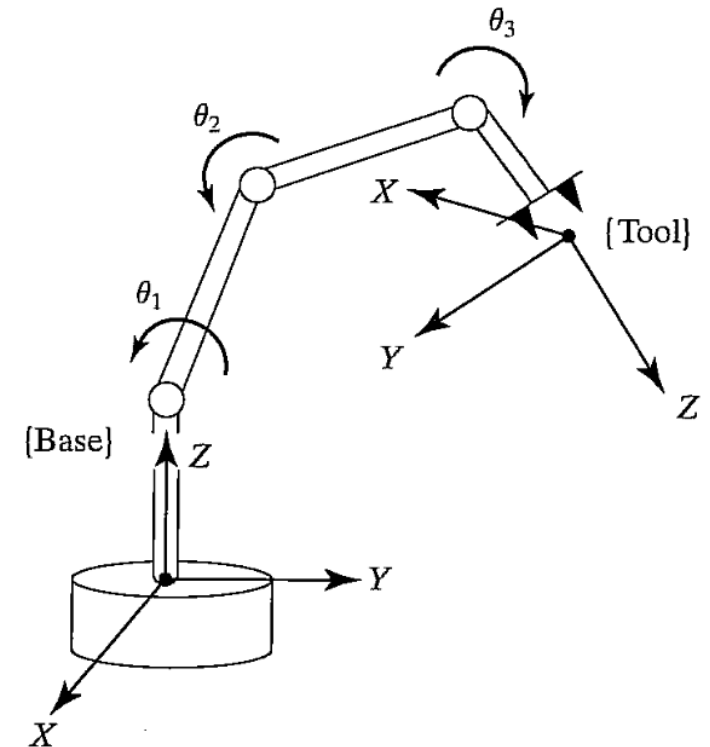
Mechanical manipulators- position and orientation

- A **coordinate system** or **frame** is rigidly attached to an object to describe its position and orientation in space.
- The position and orientation of this frame are then described with respect to some **reference coordinate system**.
- We generally describe the position of a manipulator by describing the **tool frame**, which is attached to the end-effector, relative to the **base frame**, which is attached to the non-moving base of the manipulator.



Mechanical manipulators- kinematics

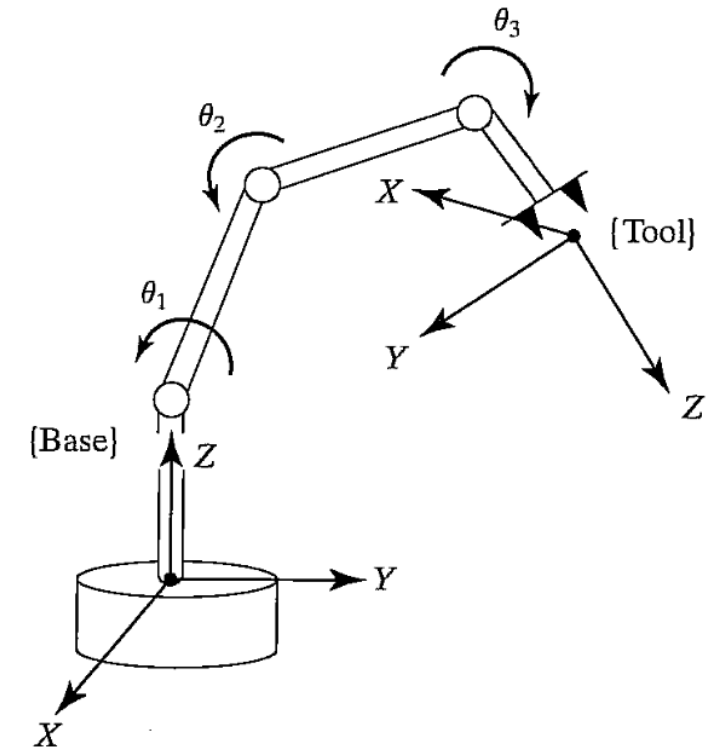
- **Kinematics** is the science of motion that treats motion without regard to the forces that cause it.
- **Within kinematics**, one studies position, velocity, acceleration, and all higher-order derivatives of the position variables.
- **Forward kinematics**
 - It is the static geometrical problem of computing the position and orientation of the end-effector.
 - Specifically, given a set of joint angles, **the forward kinematic problem is to compute the position and orientation of the tool frame relative to the base frame.**
 - Sometimes, we think of this as changing the representation of the manipulator position **from a joint space description into a Cartesian space description.**



Mechanical manipulators- kinematics

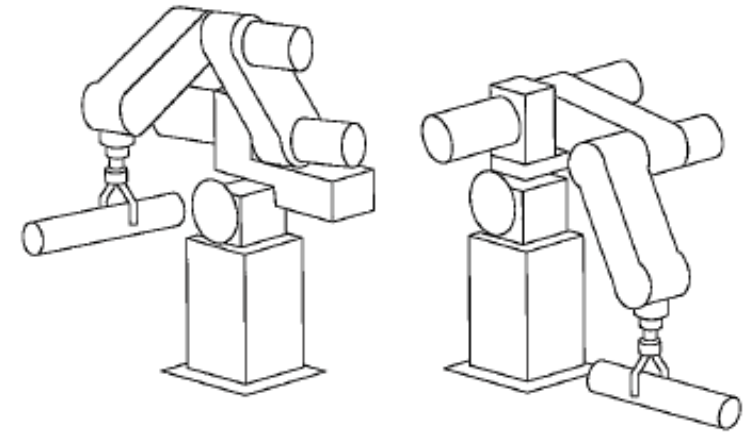
- **Inverse kinematics**

- Given the position and orientation of the end-effector of the manipulator, inverse kinematics **calculates all possible sets of joint angles** that could be used to attain this given position and orientation.
- This is a fundamental problem in the **practical use** of manipulators.
- We can think of this problem as a mapping of "locations" in **3-D Cartesian space to "locations" in the robot's internal joint space**.
- This need **naturally arises** anytime a goal is specified in external 3-D space coordinates.



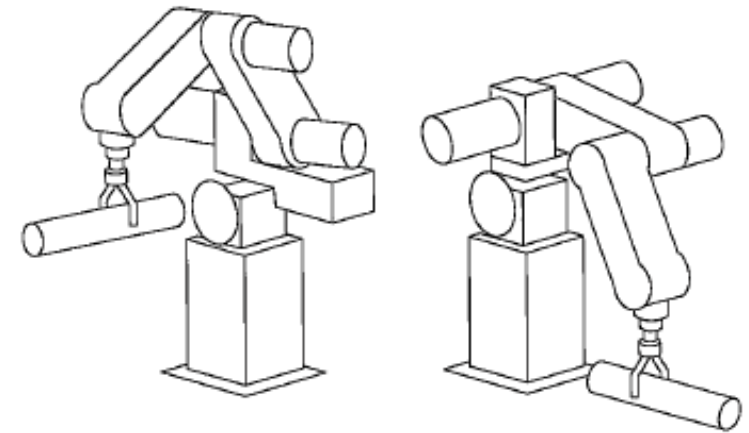
Trajectory generation- general considerations

- **Trajectory** refers to a **time history of position, velocity, and acceleration** for each degree of freedom.
- A common way of causing a manipulator to move from here to there in a smooth, controlled fashion is to cause each joint to move as specified by a **smooth function of time**.
- For our purposes, we will define a **smooth function** as a **continuous function with a continuous first derivative**.
- Commonly, each joint **starts and ends its motion at the same time**, so that the manipulator's motion appears coordinated.
- Exactly how to compute these motion functions is the problem of **trajectory generation**.



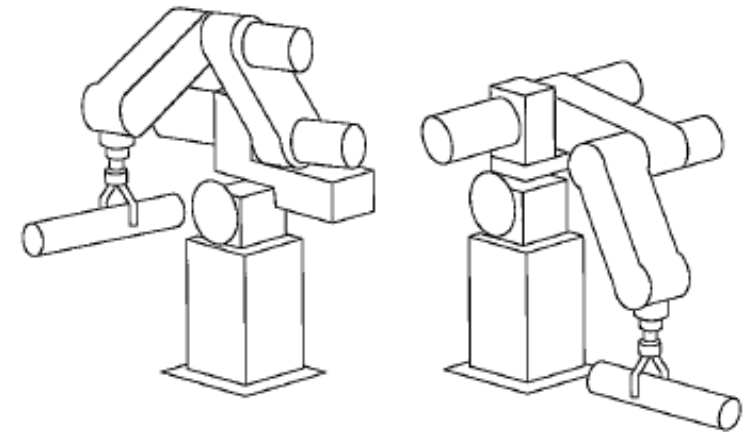
Trajectory generation- general considerations

- A **path** is described not only by a desired destination but also by some **intermediate locations**, or **via points**, through which the manipulator must pass en route to the destination.
- Thus, in completing the motion, the tool frame must pass through **a set of intermediate positions** and orientations as described by the via points.
- In such instances, the term **spline** is sometimes used to refer to **a smooth function** that passes through a set of via points.
- The name “**path points**” includes all the via points plus the initial and final points.



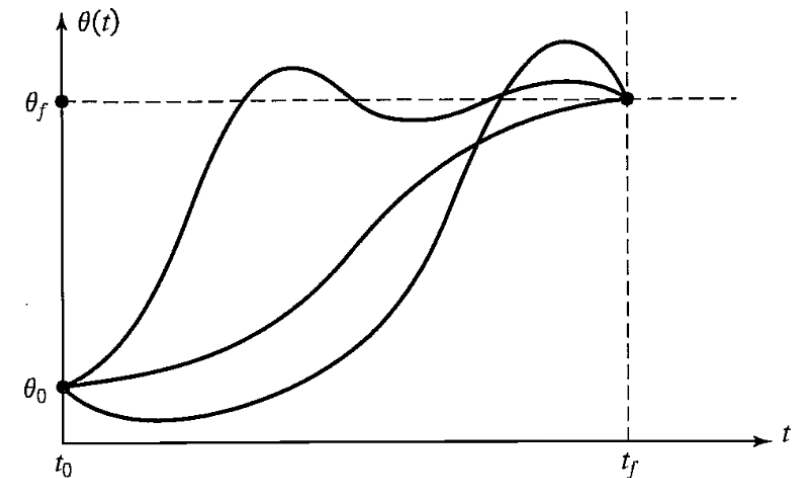
Trajectory generation: Joint-space scheme

- Each path point is usually specified in terms of a desired position and orientation of the **tool frame, $\{T\}$** , relative to the **station frame, $\{S\}$** .
- Each of the via points is "converted" into a set of desired joint angles by applying **inverse kinematics**.
- Then a **smooth function** is found for each of the **n joints** that pass through the via points and end at the goal point.
- **The time required** for each segment is the same for each joint, so that all joints will reach the via point at the same time.
- Note that, in general, this motion involves both a **change in orientation** and a **change in the position** of the tool relative to the station.



Trajectory generation: Joint-space scheme

- Consider the problem of moving the tool from its initial position to a goal position in a certain amount of time.
- The **initial position** of the manipulator is known in the form of a **set of joint angles**.
- **Inverse kinematics** allows the set of joint angles that correspond to the **goal position and orientation** to be calculated.
- **What is required is a function for each joint** whose value at t_0 is the initial position of the joint and whose value at t_f is the desired goal position of that joint.
- There are **many smooth functions**, $\theta(t)$, that might be used to interpolate the joint value.



Trajectory generation: Joint-space scheme

- In making a single smooth motion, **at least four constraints** on $\theta(t)$ are evident.
 - Two constraints on the function's value come from the selection of **initial and final values**:

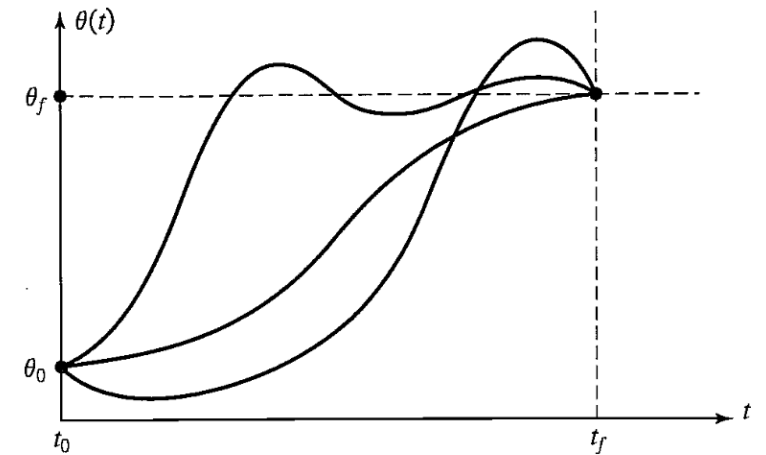
$$\theta(0) = \theta_0,$$

$$\theta(t_f) = \theta_f.$$

- Additional two constraints are **zero initial and final velocities**:

$$\dot{\theta}(0) = 0,$$

$$\dot{\theta}(t_f) = 0.$$



Trajectory generation: Joint-space scheme

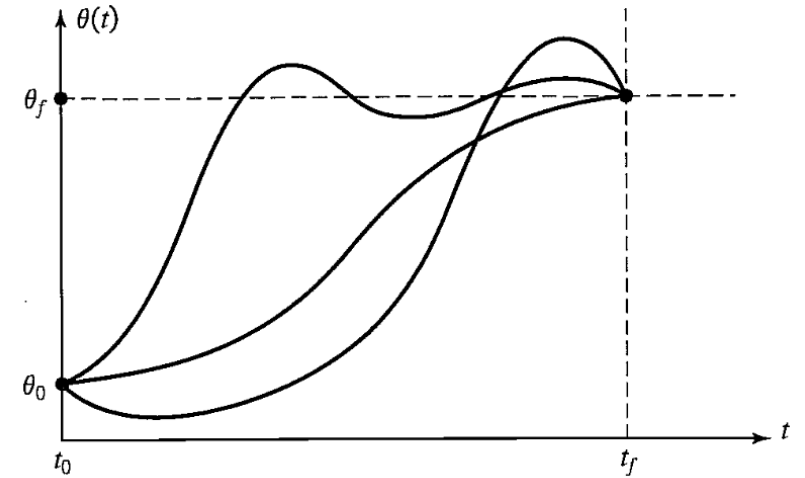
- Cubic polynomials
 - These four constraints **uniquely specify a particular cubic**.
 - A cubic has the form:

$$\theta(t) = a_0 + a_1t + a_2t^2 + a_3t^3,$$

- The joint velocities and acceleration along this path are:

$$\dot{\theta}(t) = a_1 + 2a_2t + 3a_3t^2,$$

$$\ddot{\theta}(t) = 2a_2 + 6a_3t.$$



Known condition:

$$\left\{ \begin{array}{l} \theta(0) = \theta_0, \\ \theta(t_f) = \theta_f, \\ \dot{\theta}(0) = 0, \\ \dot{\theta}(t_f) = 0. \end{array} \right.$$

Trajectory generation: Joint-space scheme

- Cubic polynomials

- Applying known conditions:

$$\theta_0 = a_0,$$

$$\theta_f = a_0 + a_1 t_f + a_2 t_f^2 + a_3 t_f^3,$$

$$0 = a_1,$$

$$0 = a_1 + 2a_2 t_f + 3a_3 t_f^2.$$

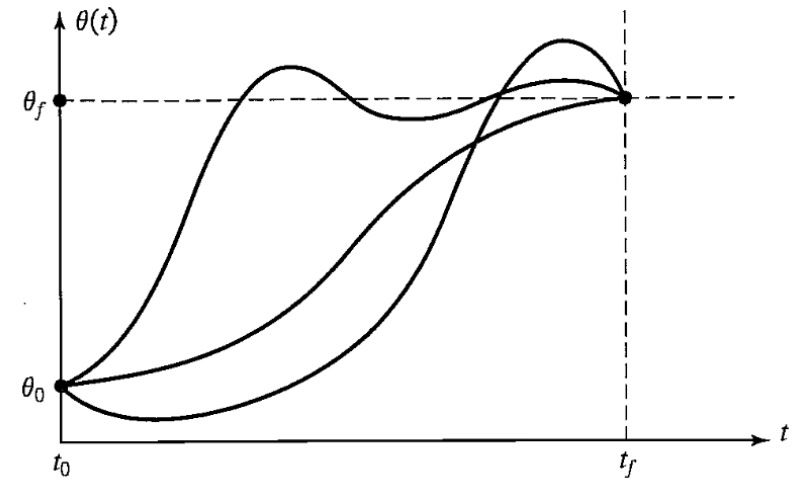
- Solving the above equations:

$$a_0 = \theta_0,$$

$$a_1 = 0,$$

$$a_2 = \frac{3}{t_f^2}(\theta_f - \theta_0),$$

$$a_3 = -\frac{2}{t_f^3}(\theta_f - \theta_0).$$



Trajectory generation: Joint-space scheme

- Cubic polynomials
 - **Example problem 1:** A single-link robot with a rotary joint is motionless at $\theta = 15$ degrees. It is desired to move the joint in a smooth manner to $\theta = 75$ degrees in 3 seconds. Find the coefficients of a cubic that accomplishes this motion and brings the manipulator to rest at the goal. Plot the position, velocity, and acceleration of the joint as a function of time.

Solution:

$$a_0 = 15.0,$$

$$a_1 = 0.0,$$

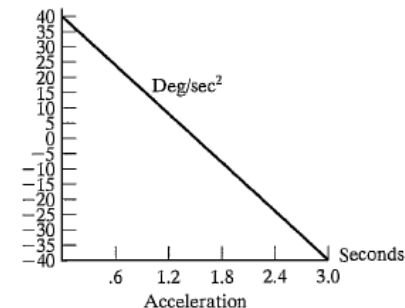
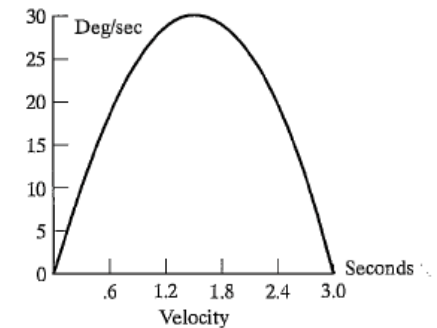
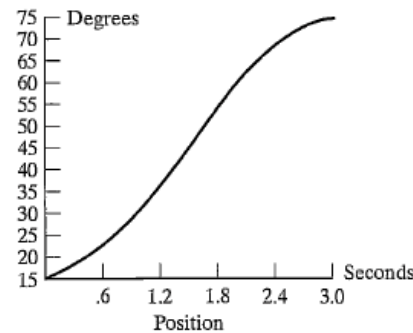
$$a_2 = 20.0,$$

$$a_3 = -4.44.$$

$$\theta(t) = 15.0 + 20.0t^2 - 4.44t^3,$$

$$\dot{\theta}(t) = 40.0t - 13.33t^2,$$

$$\ddot{\theta}(t) = 40.0 - 26.66t.$$



Trajectory generation: Joint-space scheme

- Cubic polynomials
 - **Exercise problem 1:** A single-link robot with a rotary joint is motionless at $\theta = -5^\circ$. It is desired to move the joint in a smooth manner to $\theta = 80^\circ$ in 4 seconds. Find the coefficients of a cubic that accomplishes this motion and brings the arm to rest at the goal. Plot the position, velocity, and acceleration of the joint as a function of time.

Trajectory generation: Joint-space scheme

- Cubic polynomials for a path with via points
 - Usually, we wish to be able to **pass through a via point without stopping**, and so we need to generalize the way in which we fit cubics to the path constraints.
 - As in the case of a single goal point, **each via point is usually specified** in terms of a desired position and orientation of the tool frame relative to the station frame.
 - Each of these via points is "**converted**" into a set of desired joint angles by the application of inverse kinematics.
 - We then consider the problem of **computing cubics** that connect the via-point values for each joint together in a smooth way.

Trajectory generation: Joint-space scheme

- Cubic polynomials for a path with via points
 - Now the velocity constraints at each end are not zero, but rather, some known velocity. The constraints become

$$\theta(t) = a_0 + a_1t + a_2t^2 + a_3t^3,$$

$$\dot{\theta}(t) = a_1 + 2a_2t + 3a_3t^2,$$

Insert



$$\theta(0) = \theta_0, \quad \dot{\theta}(0) = \dot{\theta}_0,$$

$$\theta(t_f) = \theta_f, \quad \dot{\theta}(t_f) = \dot{\theta}_f.$$



$$\theta_0 = a_0,$$

$$\theta_f = a_0 + a_1t_f + a_2t_f^2 + a_3t_f^3,$$

$$\dot{\theta}_0 = a_1,$$

$$\dot{\theta}_f = a_1 + 2a_2t_f + 3a_3t_f^2.$$

Solve



$$a_0 = \theta_0,$$

$$a_1 = \dot{\theta}_0,$$

$$a_2 = \frac{3}{t_f^2}(\theta_f - \theta_0) - \frac{2}{t_f}\dot{\theta}_0 - \frac{1}{t_f}\dot{\theta}_f,$$

$$a_3 = -\frac{2}{t_f^3}(\theta_f - \theta_0) + \frac{1}{t_f^2}(\dot{\theta}_f + \dot{\theta}_0).$$

Trajectory generation: Joint-space scheme

- Cubic polynomials for a path with via points
 - **Exercise problem 2:** Sketch graphs of position, velocity, and acceleration for a two-segment spline where each segment is a cubic. Sketch them for a joint where $\theta_o = 5.0^\circ$ for the initial point, $\theta_v = 15.0^\circ$ is a via point, and $\theta_g = 40.0^\circ$ is the goal point. Assume that each segment has a duration of 1.0 second and that the velocity at the via point is to be 17.5 degrees/second.

Trajectory generation: Joint-space scheme

- Cubic polynomials for a path with via points
 - **If we have the desired joint velocities at each via point**, then we simply apply the derived equations to each segment to find the required cubics.
 - There are several ways in which the desired velocity at the via points might be specified:
 1. **The user specifies the desired velocity at each via point** in terms of a Cartesian linear and angular velocity of the tool frame at that instant.
 2. **The system automatically chooses the velocities at the via points by applying a suitable heuristic** in either Cartesian space or joint space.
 3. **The system automatically chooses the velocities at the via points** in such a way as to cause **the acceleration at the via points to be continuous**.

Trajectory generation: Joint-space scheme

- Cubic polynomials for a path with via points
 - **Example derivation:** Solve for the coefficients of two cubics that are connected in a two-segment spline with continuous acceleration at the intermediate via point. The initial angle is θ_0 , the via point is θ_v , and the goal point is θ_g .
 - **Solution:**

The first cubic is

$$\theta(t) = a_{10} + a_{11}t + a_{12}t^2 + a_{13}t^3, \quad (7.12)$$

and the second is

$$\theta(t) = a_{20} + a_{21}t + a_{22}t^2 + a_{23}t^3. \quad (7.13)$$

Each cubic will be evaluated over an interval starting at $t = 0$ and ending at $t = t_{fi}$, where $i = 1$ or $i = 2$.

Trajectory generation: Joint-space scheme

- Cubic polynomials for a path with via points
 - **Example derivation:** Solve for the coefficients of two cubics that are connected in a two-segment spline with continuous acceleration at the intermediate via point. The initial angle is θ_0 the via point is θ_v and the goal point is θ_g .

Solution:

The constraints we wish to enforce are

$$\theta_0 = a_{10},$$

$$\theta_v = a_{10} + a_{11}t_{f1} + a_{12}t_{f1}^2 + a_{13}t_{f1}^3,$$

$$\theta_v = a_{20},$$

$$\theta_g = a_{20} + a_{21}t_{f2} + a_{22}t_{f2}^2 + a_{23}t_{f2}^3,$$

$$0 = a_{11},$$

$$0 = a_{21} + 2a_{22}t_{f2} + 3a_{23}t_{f2}^2,$$

$$a_{11} + 2a_{12}t_{f1} + 3a_{13}t_{f1}^2 = a_{21},$$

$$2a_{12} + 6a_{13}t_{f1} = 2a_{22}.$$

Trajectory generation: Joint-space scheme

- Cubic polynomials for a path with via points
 - **Example derivation:** Solve for the coefficients of two cubics that are connected in a two-segment spline with continuous acceleration at the intermediate via point. The initial angle is θ_0 the via point is θ_v and the goal point is θ_g .

Solution:

- These constraints specify a linear equation problem having **eight equations and eight unknowns**.
- Solving for the case $t_{f1} = t_{f2} = t_f$ we obtain

$$a_{10} = \theta_0,$$

$$a_{11} = 0,$$

$$a_{12} = \frac{12\theta_v - 3\theta_g - 9\theta_0}{4t_f^2},$$

$$a_{13} = \frac{-8\theta_v + 3\theta_g + 5\theta_0}{4t_f^3},$$

$$a_{20} = \theta_v,$$

$$a_{21} = \frac{3\theta_g - 3\theta_0}{4t_f},$$

$$a_{22} = \frac{-12\theta_v + 6\theta_g + 6\theta_0}{4t_f^2},$$

$$a_{23} = \frac{8\theta_v - 5\theta_g - 3\theta_0}{4t_f^3}.$$

Trajectory generation: Joint-space scheme

- Cubic polynomials for a path with via points
 - **Exercise problem 3:** A single-link robot with a rotary joint is motionless initially. Sketch graphs of position, velocity, and acceleration for the two-segment continuous-acceleration spline. Sketch them for a joint for which $\theta_o = 5.0^\circ$, $\theta_v = 15.0^\circ$, and $\theta_g = -10.0^\circ$, and each segment lasts 2.0 seconds.

Trajectory generation: Joint-space scheme

- Higher-order polynomials
 - Higher-order polynomials are sometimes used for path segments.
 - For example, if we wish to be able to **specify the position, velocity, and acceleration** at the beginning and end of a path segment, a **quintic polynomial** is required, namely,

$$\theta(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5,$$

where the constraints are given as

$$\theta_0 = a_0,$$

$$\theta_f = a_0 + a_1t_f + a_2t_f^2 + a_3t_f^3 + a_4t_f^4 + a_5t_f^5,$$

$$\dot{\theta}_0 = a_1,$$

$$\dot{\theta}_f = a_1 + 2a_2t_f + 3a_3t_f^2 + 4a_4t_f^3 + 5a_5t_f^4,$$

$$\ddot{\theta}_0 = 2a_2,$$

$$\ddot{\theta}_f = 2a_2 + 6a_3t_f + 12a_4t_f^2 + 20a_5t_f^3.$$

Trajectory generation: Joint-space scheme

- Higher-order polynomials
 - These constraints specify a linear set of six equations with six unknowns, whose solution is-

$$a_0 = \theta_0,$$

$$a_1 = \dot{\theta}_0,$$

$$a_2 = \frac{\ddot{\theta}_0}{2},$$

$$a_3 = \frac{20\theta_f - 20\theta_0 - (8\dot{\theta}_f + 12\dot{\theta}_0)t_f - (3\ddot{\theta}_0 - \ddot{\theta}_f)t_f^2}{2t_f^3},$$

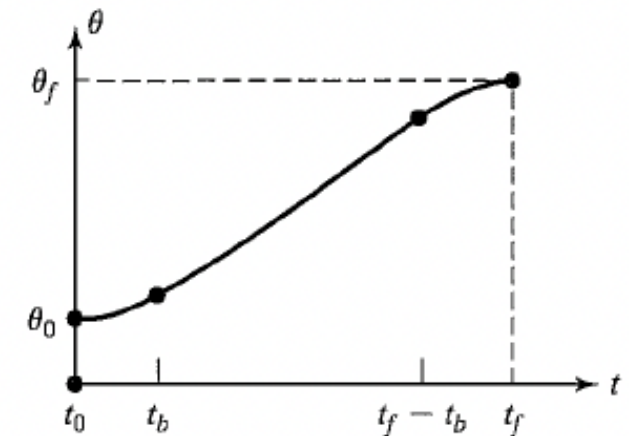
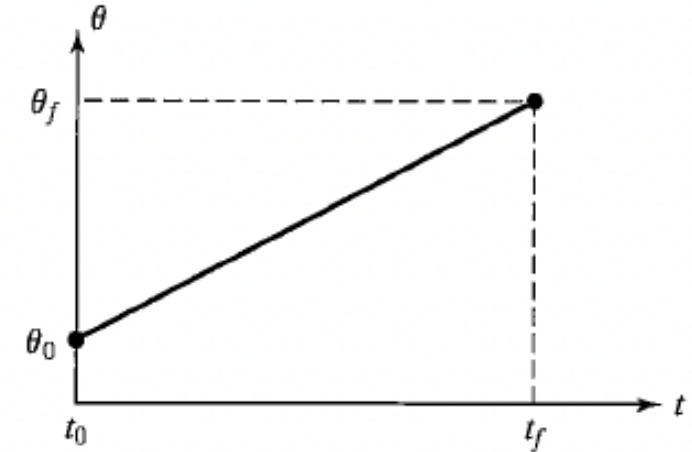
$$a_4 = \frac{30\theta_0 - 30\theta_f + (14\dot{\theta}_f + 16\dot{\theta}_0)t_f + (3\ddot{\theta}_0 - 2\ddot{\theta}_f)t_f^2}{2t_f^4},$$

$$a_5 = \frac{12\theta_f - 12\theta_0 - (6\dot{\theta}_f + 6\dot{\theta}_0)t_f - (\ddot{\theta}_0 - \ddot{\theta}_f)t_f^2}{2t_f^5}.$$

- Various algorithms are available for computing smooth functions (polynomial or otherwise) that pass through a given set of data points.

Trajectory generation: Joint-space scheme

- Linear function with parabolic blends
 - Straightforward **linear interpolation** would cause the **velocity** to be **discontinuous** at the beginning and end of the motion due to sudden jump or drop.
 - The **acceleration** will be infinite at the start and end.
 - **To create a smooth path** with continuous position and velocity, we start with the linear function but add a **parabolic blend region** at each path point.
 - **During the blend portion** of the trajectory, constant acceleration is used to change velocity smoothly, causing the linear function to be splined with two parabolic functions at two ends.



Trajectory generation: Joint-space scheme

- Linear function with parabolic blends
 - Let's assume that the **parabolic blends both have the same duration**; therefore, the same constant acceleration (modulo a sign) is used during both blends.
 - There are **many solutions** to this problem—but note that the answer is **always symmetric** about the halfway point in time, t_h , and about the halfway point in position θ_h .
 - The velocity at the end of the blend region must equal the velocity of the linear section, and so we have

$$\ddot{\theta} t_b = \frac{\theta_h - \theta_b}{t_h - t_b}, \quad \theta_b = \theta_0 + \frac{1}{2} \ddot{\theta} t_b^2.$$

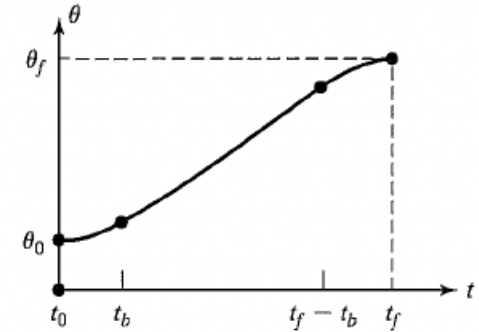


FIGURE 7.6: Linear segment with parabolic blends.

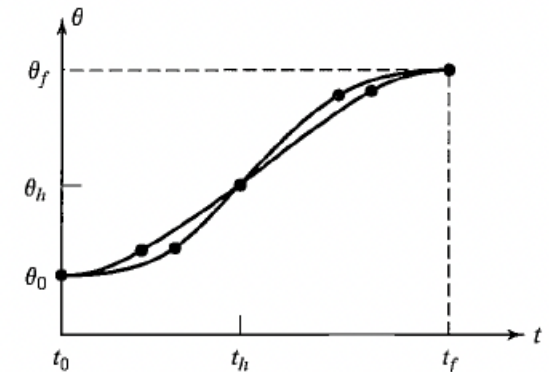


FIGURE 7.7: Linear segment with parabolic blends.

Trajectory generation: Joint-space scheme

- Linear function with parabolic blends

$$\left. \begin{aligned} \ddot{\theta} t_b &= \frac{\theta_h - \theta_b}{t_h - t_b}, \\ \theta_b &= \theta_0 + \frac{1}{2} \ddot{\theta} t_b^2. \end{aligned} \right\} \xrightarrow{t = 2t_h} \ddot{\theta} t_b^2 - \ddot{\theta} t t_b + (\theta_f - \theta_0) = 0, \xrightarrow{\text{Solving for } t_b} t_b = \frac{t}{2} - \frac{\sqrt{\ddot{\theta}^2 t^2 - 4\ddot{\theta}(\theta_f - \theta_0)}}{2\ddot{\theta}}.$$

where t is the desired duration of the motion

The constraint on the acceleration used in the blend is

$$\ddot{\theta} \geq \frac{4(\theta_f - \theta_0)}{t^2}$$

- **When equality occurs** in the above condition, the linear portion has shrunk to zero length and the path is composed of two blends that connect with equivalent slope.
- **As the acceleration used becomes larger and larger**, the length of the blend region becomes shorter and shorter.
- **In the limit**, with infinite acceleration, we are back to the simple linear-interpolation case.

Trajectory generation: Joint-space scheme

- Cubic polynomials
 - **Example problem:** A single-link robot with a rotary joint is motionless at $\theta = 15$ degrees. It is desired to move the joint in a smooth manner to $\theta = 75$ degrees in 3 seconds using a straight path with same size of parabolic blends at both ends. Calculate the straight segment velocity, blend times, and the linear time for an acceleration twice the minimum acceleration needed for this path.

Homework

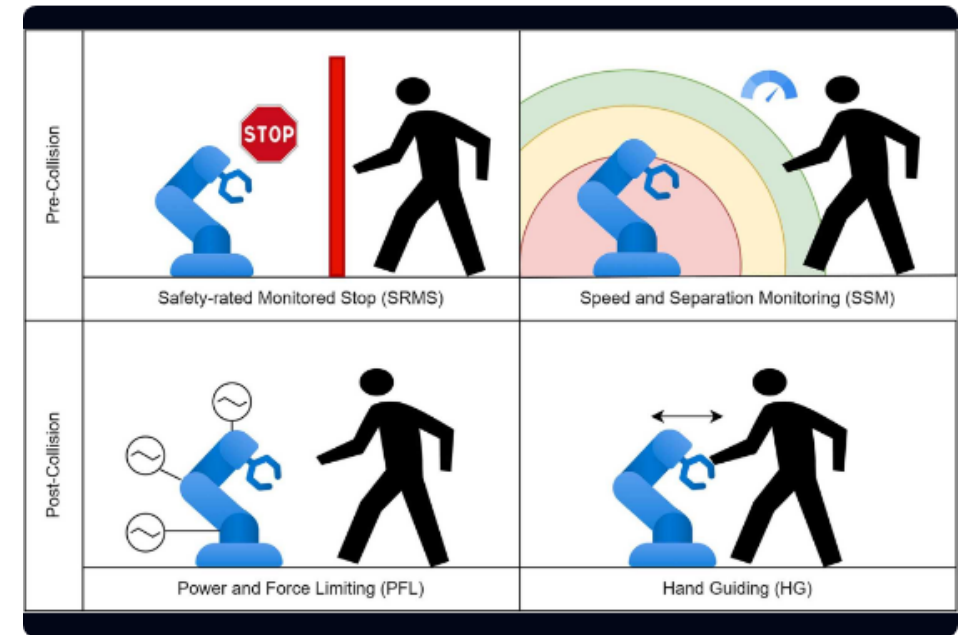
- Derive the coefficients of all the equations (cubic polynomial, cubic polynomial for path with via points, and higher-order polynomial).
- Derive the function with parabolic blends for a path with via points.
- Solve all the included exercise problems.
- Practice relevant exercise problems from the reference book.

HRI Perspectives

- The human-aware problem
 - Standard path planning algorithms often fail to produce socially acceptable motion in shared space.
 - **Why?**

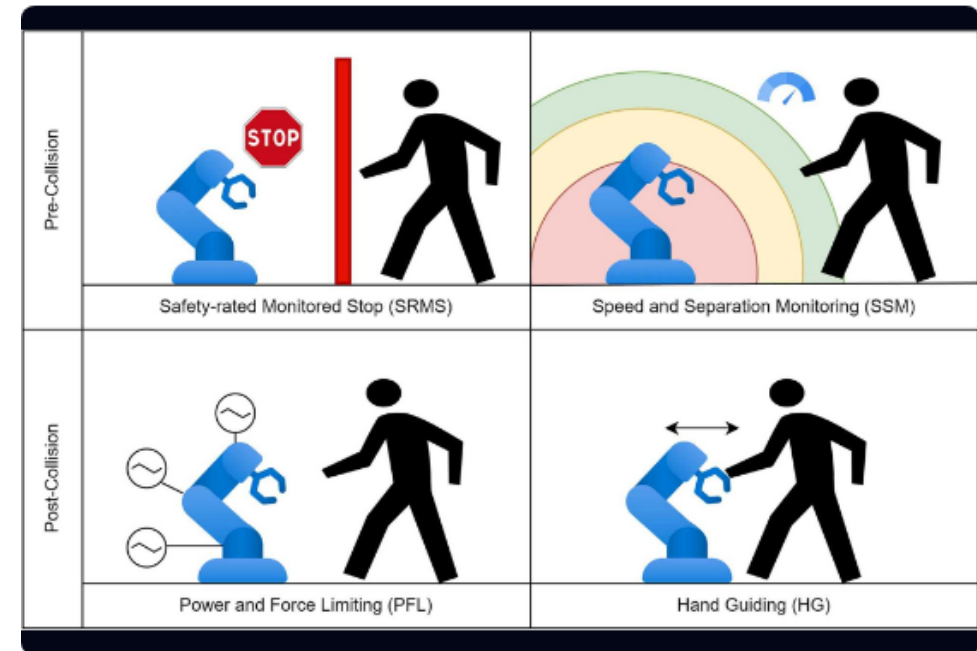
HRI Perspectives

- Active Safety & Standards
 - **ISO/TS 15066:** Defines the 4 collaborative modes.
 1. **Safety-Rated Monitored Stop (SRMS):** The robot and human share the same workspace, but never at the same time while the robot is moving. When a human steps into the collaborative zone (detected via a laser scanner or light curtain), the robot stops immediately and holds its position with its drives still powered on.
 2. **Speed & Separation Monitoring (SSM):** Trajectory velocity is dynamic, based on human distance d . The robot adjusts its speed dynamically based on how close the human is to its operating envelope.



HRI Perspectives

- Active Safety & Standards
 - **ISO/TS 15066:** Defines the 4 collaborative modes.
 3. **Power & Force Limiting (PFL):** It is the only mode where incidental physical contact between a moving robot and a human is structurally permitted. Impact energy is limited via momentum constraints.
 4. **Hand Guiding (HG):** The operator takes direct control of the robot's motion using a dedicated, hand-held control device or an intuitive end-effector handle.
 - **Safety Rating:** Real-time risk assessment overrides purely efficient paths.



HRI Perspectives

- Psychological transparency
 - **Predictability:** Does the motion match user expectations? Efficient, direct paths are typically the most predictable.
 - **Legibility:** Does the motion convey the goal early? Exaggerated motion paths help observers infer intent.
 - **Trust:** Transparent motion reduces cognitive load and "surprising" behavior, directly increasing human trust in the system.
- The personal zone
 - Humans maintain a "safety bubble" (proxemics).
 - Manipulators must respect personal space (0.45m - 1.2m) to avoid eliciting fear responses.

HRI Perspectives

- Minimizing jerk
 - Humans prefer "biological" motion profiles.
 - Trapezoidal velocity (high jerk) feels mechanical and unpredictable.
 - By using S-curve profiles or Minimum-Jerk trajectories, the robot mimics biological motion.



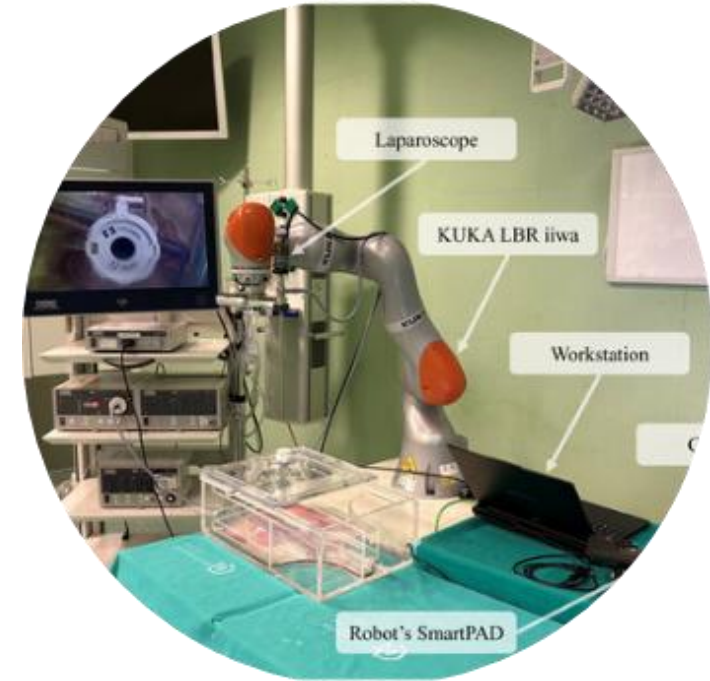
HRI Perspectives

- Collaborative handover
 - **Wait Zone:** Robot enters a passive state to let human lead.
 - **Orientation:** Tooling should be oriented for easy grasping by the human.
 - **Deceleration:** Gradual slowing as proximity decreases reduces human startle reflex.
 - **Force Sensing:** Transition from trajectory tracking to compliance upon contact.



HRI Perspectives

- Exploiting redundancy
 - **Null-space interaction:** For a 7-DOF arm, the same end-effector path can be achieved with different elbow configurations.
 - **Elbow-Down:** Generally perceived as less aggressive/threatening.
 - **Elbow-High:** May provide better visibility of the task but can invade upper proxemic zones.
 - **Motion Signaling:** Small oscillations in the null space can signal "Ready" or "Busy" states.



HRI Perspectives

- Performance metric

Metric	Description	Impact on Planning
PS (Perceived Safety)	Subjective comfort of the human user.	Informs proximity cost in optimization.
ID (Interruption Delay)	Time human pauses to process robot motion.	Decreased by increasing Legibility.
FR (Fear Response)	Physiological stress indicators (GSR, HR).	Requires smoother velocity profiles.
Fluency	Smoothness of coordinated H-R tasks.	Optimized via intent prediction models.

The End